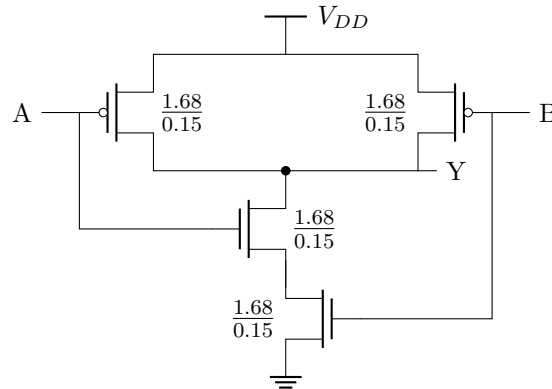
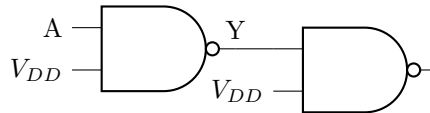


EE5311 Digital IC Design: Assignment 5
NAND gate – schematic and layout extracted simulation

1. Construct a two-input NAND gate shown below. Use appropriate number of fingers for all the devices considering the layout constraints.



The NAND gate drives another NAND gate of same size as shown below:



The input A is a pulse between 0 and V_{DD} , with a rise and fall time of 5 ps and a pulse width of 400 ps. Input B is connected to V_{DD} .

- (a) Obtain the delay of the NAND gate for $V_{DD} = 1.8\text{V}$. How does this delay compare against the one calculated using the logical effort expression?
- (b) Draw the layout containing two NAND gates connected as shown above and measure the delay using layout extracted netlist for $V_{DD} = 1.8\text{V}$.